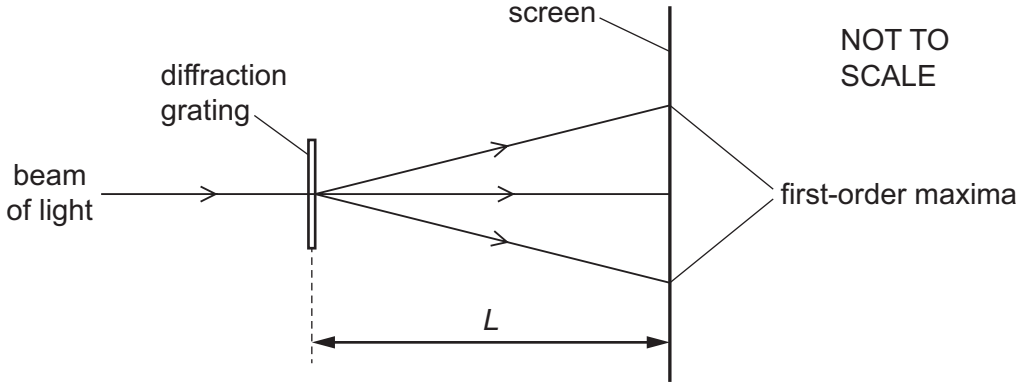
29 The diagram shows a screen that is a distance *L* from a diffraction grating. The grating has a total number of *N* lines. Any two adjacent lines are a distance *d* apart. A beam of parallel light of wavelength λ is incident normally on the grating.



Which quantities affect the distance between the first-order diffraction maxima on the screen?

	d	λ	*L*	*N*
A	✓	✓	✓	✗
B	✓	✓	✗	✗
C	✓	✗	✓	✓
D	✗	✓	✗	✓

key

✓ = affects the distance

✗ = does not affect the distance